Adaptive Poisson-Binomial Joint Conformal Intervals for Reliable Cell Counting under Distribution Shift

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Abstract

Counting cell nuclei in histopathology images is an important step toward quantitative biomarkers in computational pathology. However, the counts derived from current segmentation and detection models are typically point estimates without uncertainty quantification, a concerning limitation when the deployment data are distribution-shifted relative to the calibration data. Conformal prediction yields prediction intervals with coverage guarantees under exchangeability, but static calibration degrades under shift, while multi-class counting requires simultaneous coverage of the entire count vector. Adaptive PB-JCI Online, a lightweight online conformal calibration layer for existing cell-counting backbones, is proposed. The method models each instance's contribution with a Poisson-Binomial distribution to obtain an analytic variance scale, uses a max-statistic across classes to calibrate for joint coverage, and updates the threshold with a sliding window whose size adapts to recently observed coverage. On three datasets (PanNuke, NuInsSeg, MoNuSAC) and two backbones (SAM3, PathoSAM), the max-statistic component attains 91.7% joint coverage on MoNuSAC with $K=4$ classes. Under controlled shift on PanNuke, the method maintains near-target coverage on SAM3 and substantially improves coverage on PathoSAM, with prediction intervals narrower than those of adaptive conformal inference. In the cross-dataset shift PanNuke → NuInsSeg, where static calibration retains only about 41% empirical coverage, the adaptive mechanism restores empirical coverage to 90.0 ± 0.6% and achieves the lowest Winkler score among recent online conformal baselines.

Keywords: conformal prediction · uncertainty quantification · cell counting · distribution shift · Poisson-Binomial · histopathology